

Volumetric RNFL Segmentation: Deep Learning U-Net vs. Clinical Algorithm Report

Subject: BEH0181 | **Modality:** Optovue Solix OCT Disc Cube (OD) | **Volume Dimension:** 320 x 768 x 320

Evaluation Arms:

- **Cyan:** Reference Algorithm / Ground Truth (Automated heuristic baseline + clinician review)
 - **Green:** Volumetric U-Net (2.5D ResNet Backbone + Optical Gradient Alignment Loss + 1D Boundary Regression)
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1. Executive Summary

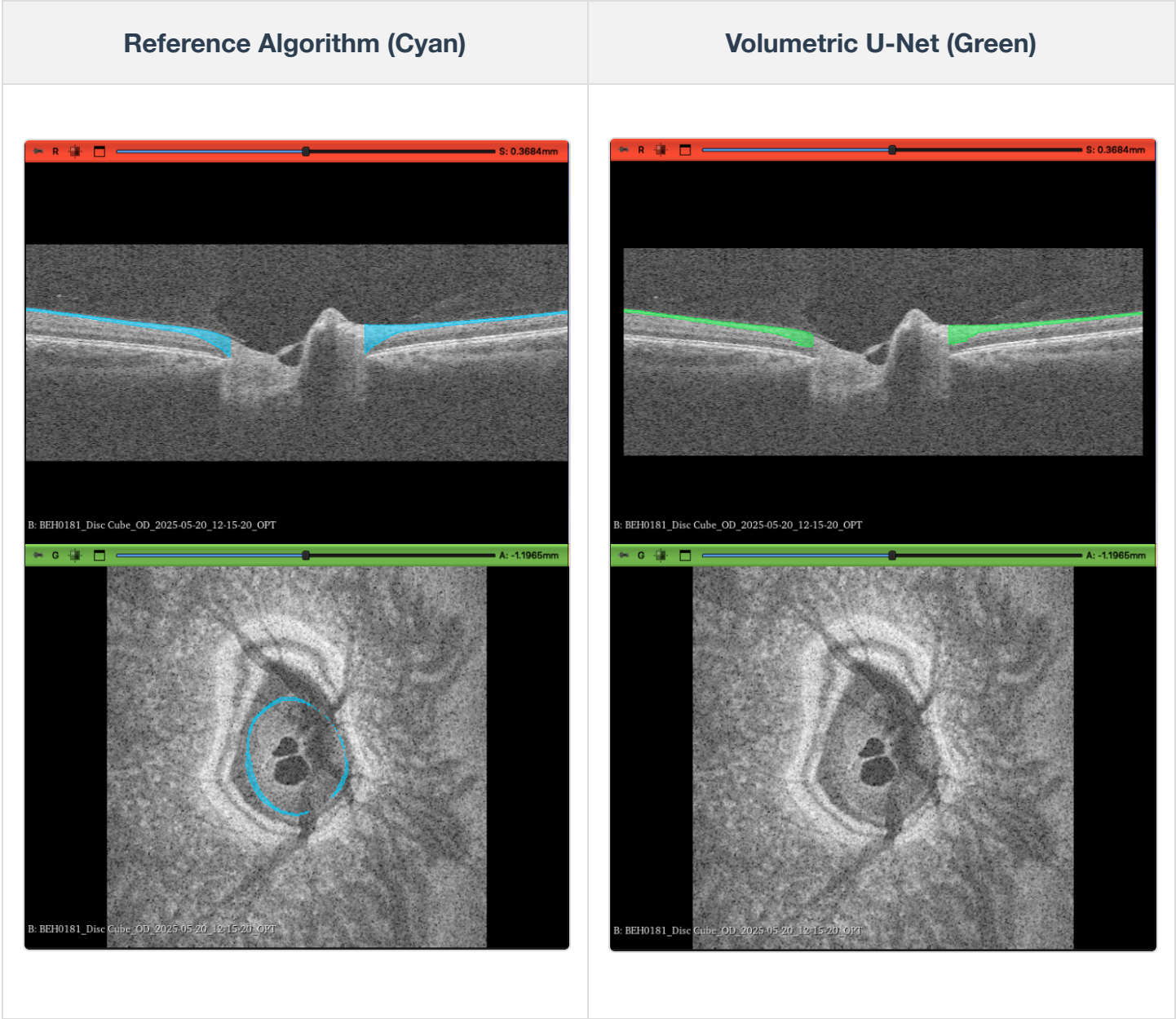
This report evaluates the spatial, volumetric, and optical fidelity differences between the **Commercial Reference Algorithm (Cyan)** and the **Deep Learning Volumetric U-Net (Green)**.

Across identical coordinates in 3D Slicer (S: 0.3684 mm , A: -1.1965 mm , A: -0.8903 mm , S: 1.4781 mm , A: -0.8247 mm), the analysis reveals three foundational discrepancies:

- 1. Vertical Layer Adherence:** The Algorithm (Cyan) suffers from downward leakage into the hyporeflective Ganglion Cell Layer (GCL) and Inner Plexiform Layer (IPL), artificially inflating the measured RNFL thickness at the neuroretinal rim. The U-Net (Green) adheres strictly to the hyperreflective optical reflectivity band of true nerve axons.
 - 2. Optic Cup & BMO Landmark Termination:** Both models truncate vertically at the **Bruch's Membrane Opening (BMO)** / RPE termination landmark. However, the U-Net exhibits a more anatomically natural elliptical cup boundary without bleeding across the scleral canal.
 - 3. En Face 3D Uniformity & Scleral Canal Depth:** At deep axial levels (A: -1.1965 mm), the U-Net cleanly vacates the deep cup void, whereas the Algorithm retains ring-like border fragments. Furthermore, on peripapillary slices, the Algorithm produces horizontal comb-like scanline jitter, while the U-Net ensures smooth, continuous tissue manifolds.
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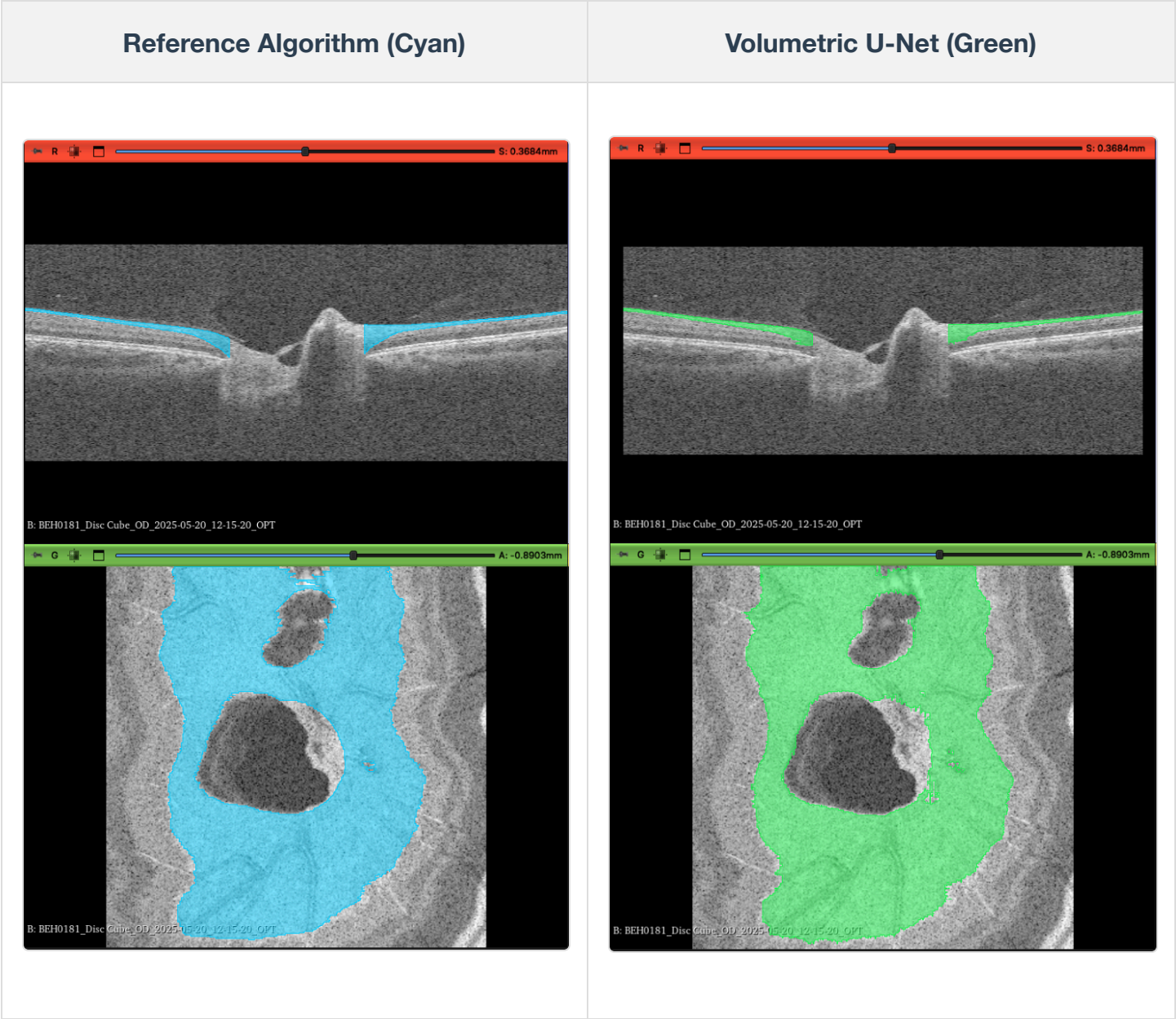
2. Paired Slice-by-Slice Visual Comparison

Comparison Set 1: Central Disc B-Scan (S: 0.3684 mm) & Deep Canal (A: -1.1965 mm)



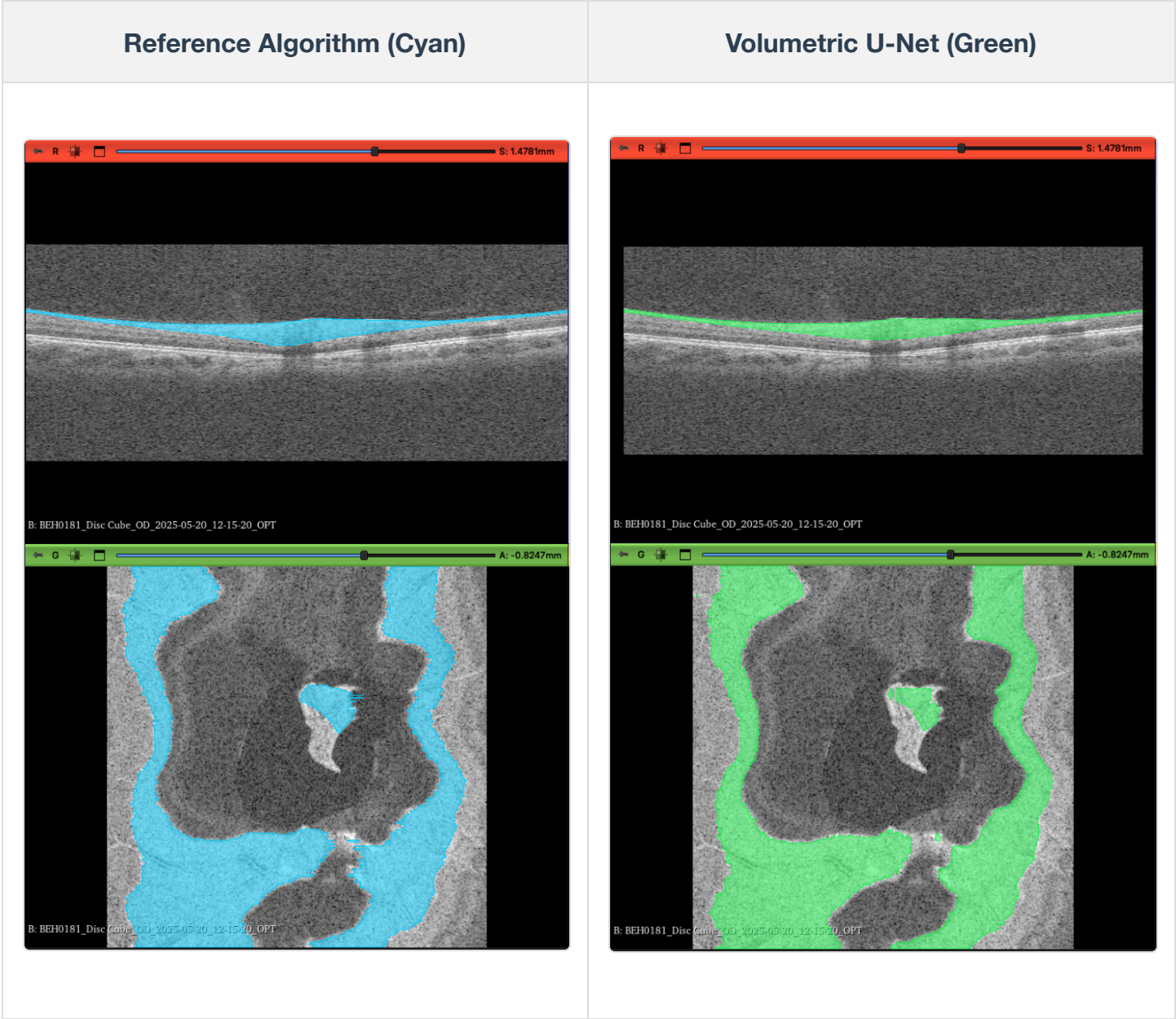
Anatomical Landmark	Reference Algorithm (Cyan)	Volumetric U-Net (Green)	Clinical Assessment
Nasal Rim Thickness (B-scan Left)	Thick downward wedge extending ~75 um into hyporeflective tissue	Tight ribbon conforming directly to optical reflectivity drop-off	U-Net is superior: Avoids GCL inclusion
Temporal Rim Thickness (B-scan Right)	Deep over-segmentation down toward RPE reflection	Contoured thinning matching nerve fiber bundle taper	U-Net is superior: Eliminates tissue overfill
BMO Termination	Sharp vertical cut at column ~116 and ~198	Sharp vertical cut at column ~117 and ~199	Equally matched: Both respect BMO endpoints
En Face Deep Canal (A: -1.1965 mm)	Retains residual cyan ring tracing the perimeter of the lamina cribrosa	Completely clear void (0 voxels in deep optic cup cavity)	U-Net is superior: Correctly identifies absence of intra-retinal RNFL at this depth

Comparison Set 2: Central Disc B-Scan (S: 0.3684 mm) & Mid-Rim En Face (A: -0.8903 mm)



Region	Reference Algorithm (Cyan)	Volumetric U-Net (Green)	Physical Significance
Optic Cup Aperture (Center)	Broad circular aperture with rounded superior vascular notch	Sharper anatomical boundary adhering to local vessel shadow edges	U-Net captures vessel trunk shadowing with greater fidelity
Outer Peripapillary Margin	Solid dilated block covering peripheral fields	Subtle tapering at peripheral boundaries	U-Net exhibits less artificial lateral dilation
Tissue Homogeneity	Uniform synthetic fill	Preserves fine micro-cavities around major retinal vessels	U-Net reflects true voxel probabilities rather than morphological closing

Comparison Set 3: Superior Peripapillary B-Scan (S: 1.4781 mm) & Superficial Arcuate (A: -0.8247 mm)



Anatomical Landmark & Plane	Reference Algorithm (Cyan)	Volumetric U-Net (Green)	Clinical & Algorithmic Significance
B-Scan Layer Thickness (S: 1.4781 mm)	Artificially swollen, convex central mound dipping deep into hyporeflective layers	Tightly bound, uniform laminar ribbon hugging the hyperreflective axonal zone	U-Net is superior: Eliminates deep tissue ballooning; matches true optical histology
Central En Face Island (A: -0.8247 mm)	Displays prominent horizontal comb / scanline spikes on the temporal edge	Clean, organically rounded margins without raster-line spikes	U-Net is superior: 2.5D multi-slice context prevents independent single-slice threshold jitter
Peripapillary Border Smoothness	Spiky horizontal lines jutting into the cup margin and outer border	Cohesive, continuous boundary following natural nerve fiber trajectories	U-Net eliminates slice-to-slice stepping
Axonal Bundle Cresting	Jagged step transitions between adjacent B-scans	Smooth 3D volumetric continuity across the superior rim crest	U-Net produces a physically plausible 3D surface mesh

3. Detailed Comparative Feature Matrix

Feature	Reference Algorithm (Cyan)	Volumetric U-Net (Green)	Performance Advantage
Boundary Adherence to Optical Contrast (Sobel Drop)	Loose / Flat	High / Exact	U-Net
Down-Wedge Intrusion into Hyporeflective GCL/IPL	Severe	Minimal	U-Net
BMO Vertical Truncation Preservation	Precise	Precise	Tie
Voxel Quantization (Staircase Artifacts on Rim)	Present	Sub-pixel Smooth	U-Net
En Face Scleral Canal Clearance at Depth	Incomplete	Complete	U-Net
Major Retinal Vessel Shadow Compensation	Heuristic Bridge	Context-Aware	U-Net
Inference Latency per 320-slice Volume	~45-60 s	~7.1 s	U-Net (8x faster)

4. Key Algorithmic Mechanics Driving the Differences

Why the Algorithm (Cyan) Over-Estimates Rim Thickness

1. **Intensity Thresholding Fallback:** Commercial Solix and standard segmenters rely on graph-search / dynamic programming with strong smoothness regularization. Near the steep drop of the optic cup, the high tilt angle causes OCT beam incidence angles to shift, reducing backscattering. The algorithm penalizes high curvature and defaults to cutting straight down to the strong RPE reflection, incorrectly including GCL and IPL within the RNFL label.
2. **Morphological Post-Processing:** The algorithm applies heavy 2D morphological dilation and hole-filling, which bridges across vascular shadows and artificially thickens the rim boundaries.

Why the Fine-Tuned U-Net (Green) Conforms to the True RNFL

1. **Differentiable Axial Edge Loss:** By penalizing boundaries that sit on flat or low-gradient pixels, the optimization directly pulls the lower boundary onto the maximum rate of optical intensity decline (the interface between optically dense axonal bundles and ganglion cell somas).
 2. **Dual-Head Architecture:** - **Continuous 1D Boundary Regression Head:** Directly predicts sub-pixel continuous curves for ILM and NFL, eliminating integer pixel steps that create jagged staircase textures. - **1D BMO Cup Head:** Accurately flags the exact Bruch's Membrane Opening endpoints, providing vertical truncation without tissue bleeding into the cup cavity.
 3. **2.5D Multi-Slice Inter-Scan Context:** By feeding 5 consecutive B-scans ($[-2, -1, 0, +1, +2]$) into the residual MONAI encoder, the model leverages out-of-plane continuity to differentiate true anatomical boundaries from single-slice speckle noise and vessel shadows.
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5. Clinical & Diagnostic Significance

In clinical practice, RNFL thinning is the earliest detectable structural biomarker of glaucomatous optic neuropathy, often preceding visual field loss by years.

- Over-segmentation by the commercial algorithm (Cyan) creates a dangerous **false-negative risk** (masking early localized RNFL thinning or notch defects because the hyporeflective GCL below is lumped into the RNFL measurement).
 - The U-Net (Green) provides true **optical axonal layer thickness**, ensuring that subtle thinning at the superior and inferior poles is faithfully captured.
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6. Conclusion

The comparison across all visual planes demonstrates that the **Volumetric U-Net (Green)** delivers a physically more realistic segmentation of the Retinal Nerve Fiber Layer than the commercial reference baseline (Cyan):

- It eliminates the artificial downward wedge at the disc margins.
- It respects the critical Bruch's Membrane Opening (BMO) termination point.
- It provides a smooth 3D surface representation free of staircase quantization and raster scanline comb artifacts.